

MH2500 Probability and Introduction to Statistics

Slides for Week 13-2

Topics: Introduction to Statistics

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- In this lecture, we give an extremely brief introduction to statistics, covering only some parts of Chapter 8 (for confidence intervals) and Chapter 9 for hypothesis testing.
- Likelihood ratio tests and more hypothesis testing will be covered in the next course **MH3500**.
- Here we will talk about:
 - 1 Sampling.
 - 2 Confidence interval.
 - 3 Hypothesis Testing.

Sampling (Review)

Definition

Let $X_1, \dots, X_n$ be i.i.d random variables drawn from an unknown distribution of mean μ and variance σ^2 .

The **sample mean** and the **sample variance** are defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

What we already know:

$$\mathbb{E}(\bar{X}) = \mu, \quad \mathbb{E}(S^2) = \sigma^2.$$

Confidence Interval

- An **interval estimator** of a population **parameter** is a (random) interval **that is predicted to contain the parameter**.
- The **confidence** we ascribe to the interval is the **probability that it will contain the parameter**.
- We only study normal distributions of **unknown mean** and **known variance**.

Let $X_1, \dots, X_n$ be a sample of size n from a normal distribution of variance σ^2 (known) and $\bar{X}$ be the sample mean. We are trying to estimate μ ,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Since $P\{Z \leq 1.96\} = 0.95$, we have

$$P\left\{|\bar{X} - \mu| \leq 1.96 \frac{\sigma}{\sqrt{n}}\right\} = 0.95,$$

$$P\left\{\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right\} = 0.95.$$

$$P\left\{\mu \in \left[\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right]\right\} = 0.95.$$

- The interval $\left[\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$ is a 95% **confidence interval estimator** of μ .
- If the observed value of $\bar{X}$ is $\bar{x}$, then we call the interval $\left[\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right]$ a **95% confidence interval estimate** of μ .

In the long run, 95% of the interval estimates so constructed will contain the mean of the population from which the sample is drawn.

Example 1

To estimate μ , the average nicotine content of a newly marketed cigarette, 44 of these cigarettes are randomly chosen, and their nicotine contents are determined.

- (a) If the average nicotine finding is 1.74 milligrams, what is a 95% confidence interval estimator of μ ?
- (b) How large a sample is necessary for the length of the 95% confidence interval to be less than or equal to 0.3 milligrams?

Assume that it is known from past experience that the standard deviation of the nicotine content of a cigarette is equal to 0.7 milligrams.

Solution:

(a)

$$\left[1.74 - 1.96 \frac{0.7}{\sqrt{44}}, 1.74 + 1.96 \frac{0.7}{\sqrt{44}} \right] = [1.533, 1.947].$$

(b) The length of the 95% confidence interval is

$$2 \times 1.96 \times \frac{0.7}{\sqrt{n}} = \frac{2.744}{\sqrt{n}} < 0.3.$$

Hence

$$\sqrt{n} > 9.147, \quad n \geq 84.$$

Example 2

Hypokalaemia is diagnosed when the blood potassium level is below 3.5 mmol/L. The potassium in a blood sample varies from sample to sample and follows a normal distribution with unknown mean but standard deviation 0.10.

- A patient's potassium is measured over 4 days.

The sample over 4 days is 3.2, 3.9, 3.3, 4.0.

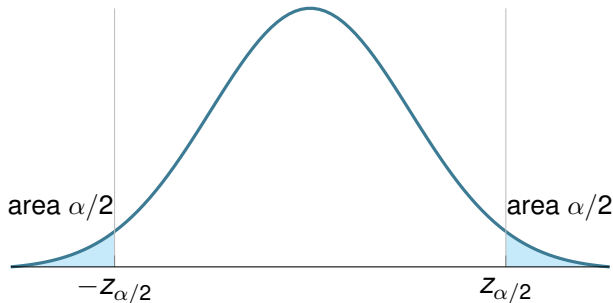
Construct a 95% confidence interval for the mean potassium and discuss whether the patient is likely to be diagnosed with hypokalaemia.

- Sample mean: 3.6
- 95% CI:

$$[3.6 - (1.96) \left(0.1/\sqrt{4}\right), 3.6 + (1.96) \left(0.1/\sqrt{4}\right)] = [3.502, 3.698]$$

In general, we can ask for $100(1 - \alpha)$ percent confidence intervals.

$$P \left\{ \bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right\} = 1 - \alpha.$$



- 90% CI:

$$z_{0.05} = 1.645$$

- 95% CI:

$$z_{0.025} = 1.96$$

- 99% CI:

$$z_{0.005} = 2.576$$

One-Sided Confidence Interval

Upper confidence bound

$$x_{\text{upper}} = \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

meaning that

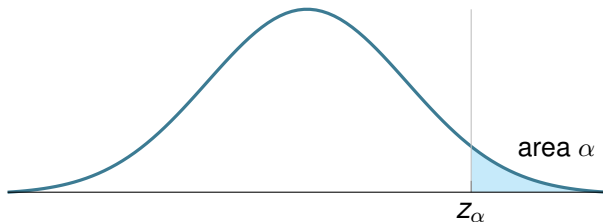
$$P\{\mu \in (-\infty, x_{\text{upper}}]\} = 1 - \alpha$$

Lower confidence bound

$$x_{\text{lower}} = \bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$$

meaning that

$$P\{\mu \in [x_{\text{lower}}, \infty)\} = 1 - \alpha$$



Hypothesis Testing – Introduction

Suppose two men are arguing about a large box of steel nails.

- A says the average is 3cm.
- B says the average is different from 3cm.

They took a sample of 16 nails and the average is 3.054cm.

Known:

the length of a nail follows a normal distribution of SD 0.1.

$$\text{(Null hypothesis)} \quad H_0 : \mu = 3$$

$$\text{(Alternative hypothesis)} \quad H_a : \mu \neq 3$$

The null hypothesis will be rejected if it appears to be inconsistent with the sample data, and will not be rejected otherwise.

Idea:

Suppose that H_0 is true.

- Look at how much $\bar{X}$ deviates from the mean μ .
- If it deviates more, it is less likely to happen.
- Note that for random samples of size n , $\bar{X} \sim N\left(\mu, \frac{0.1^2}{n}\right)$. Let

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.054 - 3}{0.1/\sqrt{16}} = 2.16.$$

This is an example of a **test statistics**.

- Tests using the z-statistic are called **z-tests**.

Let

$$p = 2P\{\bar{X} \geq 3.054\} = 2P\{Z \geq 2.16\} = 2 \cdot 0.0154 = 0.0308.$$

This number is called **observed significance level**.

- The observed significance level is the chance of getting a test statistic as extreme as, or more extreme than, the observed one. The chance is computed on the basis that H_0 is true.
- The smaller this chance is, the stronger the evidence **against H_0** .

To determine whether to accept or reject H_0 , we need a **threshold of significance level**.

- If $p \leq \text{threshold}$, reject H_0 .
- If $p > \text{threshold}$, do not reject H_0 .

Suppose that $X_1, \dots, X_n$ are drawn from a normal distribution with **unknown mean μ** and **known variance σ^2** .

Problem

Test with significance level α :

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_a : \mu \neq \mu_0.$$

Work backwards:

- From α , determine a **critical region** such that if the test statistic falls in the region, H_0 will be rejected.

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

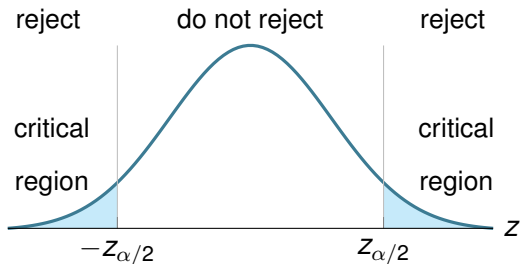
Since our H_a is $\mu \neq \mu_0$, which is **double-sided**, the critical region is of the form $\{|Z| \geq \tau\}$.

We need

$$P\{|Z| \geq \tau\} = \alpha,$$

and so

$$\tau = Z_{\alpha/2}$$

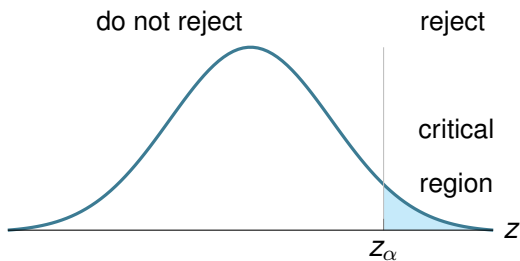


One-sided Test

Test with significance level α

$$H_0 : \mu \leq \mu_0 \quad \text{against} \quad H_a : \mu > \mu_0.$$

Critical region is $Z \geq \tau$, where $\tau = z_\alpha$.



Hypothesis Testing – Terminology

- **Null hypothesis:** H_0 .
- **Alternative hypothesis:** H_a .
- **Type I error:** Rejecting H_0 when H_0 is true.
- **Type II error:** Not rejecting H_0 when H_0 is false. The probability of a type II error is denoted by β .
- **Significance level** of the test: The probability of a type I error, denoted by α .

Hypothesis Testing – Steps

These are the steps to set up a test.

1. State the null hypothesis, H_0 .
2. State the alternative hypothesis, H_a .
3. Set the level α .
4. Collect data or examine the given data.
5. Calculate the test statistic.
6. Determine the critical region.
7. Conclusion.

Note:

$H_a : \mu < \mu_0$ and $H_a : \mu > \mu_0$ are called **one-sided alternatives**, while $H_a : \mu \neq \mu_0$ is a **two-sided alternative**.

Example

The output voltage of an electronic device manufactured by a particular company is specified to be 5 volts.

- The population standard deviation of these devices is known to be $\sigma = 0.1$ volts.

A random sample of 40 devices is selected and their output voltages recorded, giving a sample mean of 5.030 volts.

- (a) Test the hypothesis that the devices have a mean output voltage of 5 volts against the alternative test that it is more than 5 volts.
Use $\alpha = 0.05$.
- (b) What is the **type II error** β if the true mean voltage output of the devices is 5.010 volts?

(a) $H_0 : \mu = 5$, $H_a : \mu > 5$.

Test statistic z :

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{5.03 - 5}{0.1/\sqrt{40}} = 1.90.$$

- Using significance level $\alpha = 0.05$, the critical region is $z \geq 1.645$.

Conclusion: Since $z = 1.90 \geq 1.645$, we reject H_0 .

(b)

$$P\{\text{committing a type II error}\} = P\{\text{do not reject } H_0 \mid \mu = 5.010\}$$

Note that

$$\text{do not reject } H_0 \iff \frac{\bar{x} - 5}{\sigma/\sqrt{40}} < 1.645 \iff \bar{x} < 5.026.$$

Hence $P\{\text{committing a type II error}\}$

$$\begin{aligned} &= P\{\bar{x} < 5.026 \mid \mu = 5.010\} = P\left\{z < \frac{5.026 - 5.010}{0.1/\sqrt{40}}\right\} \\ &= P\{z < 1.012\} \approx 0.8438. \end{aligned}$$

Example Reconsidered

Hypokalemia is diagnosed when the blood potassium level is below 3.5 mmol/L. The potassium in a blood sample varies from sample to sample and follows a normal distribution with unknown mean but standard deviation 0.10. A patient's potassium is measured over 4 days. The sample over 4 days is

3.2, 3.9, 3.3, 4.0.

Discuss whether the patient is likely to be diagnosed with hypokalemia.

- $H_0: \mu < 3.5$, $H_a: \mu \geq 3.5$
- Test statistic: $z = (3.6 - 3.5)/(0.05) = 2$.
- We can reject H_0 even at the significance level of 2.5%.